MIL-E-1/87F
14 January 1972
SUPERSEDING
MIL-E-1/87E
7 February 1966

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 5783WA

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Diode, subminiature, voltage reference

Outline --- 3-6 (EIA)

Base --- In-line flexible leads, tinned

Envelope --- T3

Cathode --- Glow discharge

Base connections:

Pin No.	---	1	3	5
Element	---	k	a	k

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Total darkness ionization voltage	Ambient light ionization voltage	Operating voltage	Operating current	TE	Alt
Unit:	Vdc	Vdc	Vdc	mAdc	°C	ft
Maximum:	---	---	91	3.5	155	Note 1
Minimum:	140	120	81	1.5	---	---
TEST CONDITIONS:	---	---	---	---	---	---

GENERAL:

Qualification - Required

Reliable tube

(F) denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
3347	Ionization voltage (1)	2	Ebb Ib = 1.5 to 3.5 mAdc; illumination = 5 to 50 ft candle power	0.4	II	Ez	---	120	Vdc
3337	Voltage drop (1)	-	Ebb Ib = 3.5 mAdc	0.4	II	Etd	83.0	89.0	Vdc
3337	Voltage drop (2)	-	Ebb Ib = 1.5 mAdc	0.4	II	Etd	83.0	89.0	Vdc
3335	Regulation	-	Etd (1) minus Etd (2)	0.4	II	Reg	---	3.0	Vdc
3345	Voltage jump	-	Ebb Ib = 1.5 to 3.5 mAdc	0.4	II	Jump	---	5.0	mVdc
1201	Short and discontinuity detection	-		0.4	II	---	---	---	---
	Quality conformance inspection, part 2								
3347	Ionization voltage (2)	3		6.5	Code G	Ez	---	140	Vdc
3305	Leakage current	-	Eb = 50 Vdc; Rp = 3,000 ohms	6.5	Code G	LIb	---	20	µAdc
3337	Voltage drop (3)	-	Ebb Ib = 2.5 mAdc	6.5	Code G	Etd	83	88	Vdc
3278	Noise	-	Ebb Ib = 3.5 mAdc	1.0	I	EB	---	20	mVac
---	Repeatability	4	Ebb Ib = 2.5 mAdc	6.5	Code E	Etd	---	100	mVdc
1031	Low-frequency vibration	-	Rp = 10,000 ohms; Ebb Ib = 2.5 mAdc; F = 40 Hz; 15 G	6.5	Code G	Ep	---	50	mVac
1116	Lead fatigue	-		2.5	Code G	---	---	---	---
1041	Shock	-	450 G	---	---	---	---	---	---
1031	Vibration fatigue	5	2.5 G; fixed frequency; F = 25 Hz to 60 Hz	6.5	---	---	---	---	---
---	Post-shock and vibration-fatigue test end points:								
1031	Low-frequency vibration	-		---	---	Ep	---	100	mVac
3347	Ionization voltage (1)	-		---	---	Ez	---	120	Vdc
3337	Voltage drop (1)	-		---	---	Etd	81	91	Vdc
3337	Voltage drop (2)	-		---	---	Etd	81	91	Vdc
3335	Regulation	-		---	---	Reg	---	4.0	Vdc
2126	Glass strain	-		2.5	I	---	---	---	---
1105	Permanence of marking	-		---	---	---	---	---	---

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL PERCENT EFFECTIVE	INSPECTION LEVEL OR CLASS	ALLOWABLE DEFECTIVES PER CHARACTERISTIC SAMPLE	ALLOWABLE DEFECTIVES PER CHARACTERISTIC CUMULATIVE SAMPLE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 3										
(F) 1516	Stability life	-	Ebb Ib = 2.5 mAdc; TA = room	---	---	---	---	---	---	---	---
---	Stability life-test end point:										
---	Change in voltage drop (3) of individual tubes	-		---	---	---	---	ΔEtd t	---	200	mVdc
1521	Survival-rate life	-	Stability life-test, or equivalent conditions	---	---	---	---	---	---	---	---
---	Survival-rate life-test end points:										
1201	Short and discontinuity detection	-		---	---	---	---	---	---	---	---
---	Change in voltage drop (3) of individual tubes			---	---	---	---	ΔEtd t	---	1.0	Vdc
1501	Intermittent life	6	Stability life-test conditions: TE = 155°C (min)	---	---	---	---	---	---	---	---
---	Intermittent life-test end points (500 hours):										
---	Inoperatives	-		---	---	1	3	---	---	---	---
3335	Regulation	-		---	---	1	3	Reg	---	4.0	Vdc
3337	Voltage drop (1)	-		---	---	1	3	Etd	81	91	Vdc
3337	Voltage drop (2)	-		---	---	1	3	Etd	81	91	Vdc
3337	Voltage drop (3)	-		---	---	1	3	Etd	83	88	Vdc
---	Change in voltage drop (3) of individual tubes	-		---	---	1	3	ΔEtd t	---	4	Vdc
3347	Ionization voltage (1)	-		---	---	1	3	Ez	---	120	Vdc
---	Total defectives	-		---	---	4	8	---	---	---	---

NOTES:

1. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage in the basic document.
2. This test to be performed at the conclusion of the holding period.
3. Conditions for this test shall be those of ionization voltage (1), except testing shall be done in total darkness and the tube shall not have conducted or have been exposed to light for at least 24 hours prior to testing.

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NOTES: -Continued

4. Repeatability shall be defined as the maximum shift in tube voltage drop during repeated firings of the tube. The tube shall be tested in the following manner:

   (a) Etd shall be read at a 2.5 mAdc drain.

   (b) The tube shall be turned off for 1 minute.

   (c) The tube shall be restarted and operated at the same current.

   (d) Etd shall be read after 1 minute of operation.

   (e) The "on-off" cycle shall be repeated a minimum of five times. The maximum difference in Etd shall be taken as the measure of repeatability.

(F) 5. This test shall be conducted on the initial lot and thereafter on a lot approximately every 6 months. When one lot has passed, the 6-month rule shall apply. In the event of lot failure, the lot shall be rejected and the succeeding lots shall be subjected to this test until a lot passes. MIL-STD-105, sample size code letter E, shall apply.

(F) 6. Envelope temperature (TE) requirements, when measured in accordance with the temperature by conduction-band measurement (method 1226), will be satisfied if a tube having bogey tube drop (Etd = ±3 percent) under normal test conditions, is determined to operate at or above minimum specified temperature at any position in the life-test rack.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Air Force - 11, 17
DSA - ES
ARMY - MI

User activities:
Army - WC
Navy - AS, OS, MC, CG, SH

Preparing activity: Navy - EC

Agent: DSA - ES

(Project 5960-2593)

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NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-26254

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